Final Project Report: Electric Vehicle Market Performance & Insights

A Comprehensive Business Intelligence & Analytical Framework for Global EV Market Trends

1. INTRODUCTION

1.1 Project Overview

This project involves designing and developing an interactive Business Intelligence (BI) dashboard using Tableau to analyze the global Electric Vehicle (EV) market. It visualizes key metrics such as total sales, battery chemistry efficiency, pricing trends, and geographical production footprints.

1.2 Purpose

The purpose is to provide automotive strategy executives and market analysts with a centralized, data-driven tool to make informed decisions regarding supply chain investments, battery platform development, and regional market targeting.

2. IDEATION PHASE

2.1 Problem Statement

Executives and supply chain managers struggle with scattered, static reports, making it difficult to correlate EV battery chemistries with market pricing, range, and global sales. This fragmentation risks poor strategic platform decisions.

2.2 Empathy Map Canvas

- **Target User:** Automotive Strategy Executive / EV Market Analyst.
- **Pains:** Frustrated by scattered data, fear of investing massive capital into the wrong battery platform, and unseen supply chain risks.
- **Gains:** Needs a unified, interactive Tableau dashboard for instant insights and visual proof of price vs. range to confidently present recommendations.

2.3 Brainstorming

The solution conceptualized was a 3×3 interactive dashboard accompanied by a 6-point Tableau Story, driven by a raw CSV/Excel dataset. The focus was on regional adoption, battery types (NMC, LFP, Sodium-Ion), and segment pricing.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey involves three distinct stages:

1. **Access & Discovery:** Opening the Tableau Public link.
2. **Exploring & Filtering:** Drilling down by region or battery type using interactive charts.
3. **Extracting Insights:** Navigating the Tableau Story tabs and exporting reports for executive presentations.

3.2 Solution Requirements

Requirement Type	Description
Functional	Data ingestion, cleaning, KPI calculation, dynamic dashboard generation with multi-faceted filtering.
Non-Functional	Usability optimized via Tableau typography, sub-5-second render latency, public data security, and 24/7 web availability.

3.3 Data Flow Diagram

The operational data pipeline is structured as follows:

Raw Data (CSV) → Tableau Data Connection (Cleaning/Formatting) → Tableau Local Extract (.twbx) → Tableau Desktop (Visualization) → Tableau Public Server (Hosting) → End-User Web Dashboard

3.4 Technology Stack

Component	Selected Technology	Purpose
BI Tool	Tableau Desktop Public Edition	Visual interface and dashboard construction
Hosting	Tableau Public	Cloud distribution and interactive web access
Data Source	CSV / Excel Datasets	Baseline operational and market data
Logic Engine	Tableau Calculated Fields	Dynamic metric and KPI derivation

4. PROJECT DESIGN

4.1 Problem Solution Fit

The fragmented data problem is solved by consolidating global EV market data into a single, interactive 9-visual Tableau Dashboard that visually maps sales, pricing, and battery chemistries geographically.

4.2 Proposed Solution

To ingest raw EV datasets into Tableau, clean the data using calculated fields, and build a cohesive analytical experience featuring real-time KPI scorecards, interactive filtering, and geographical production mapping.

4.3 Solution Architecture

Data flows from raw CSV/Excel files into Tableau Desktop for ingestion and transformation, is stored in highly optimized Tableau Data Extracts (.twbx), rendered as a 3×3 dashboard, and published to the Tableau Public Cloud Server for end-user interaction via a web browser.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Executed over 3 sprints. Sprint 1 focused on data access and KPI calculation. Sprint 2 focused on geospatial mapping and interactive filtering. Sprint 3 focused on business insights storytelling and Tableau Public deployment.

6. FUNCTIONAL AND PERFORMANCE TESTING

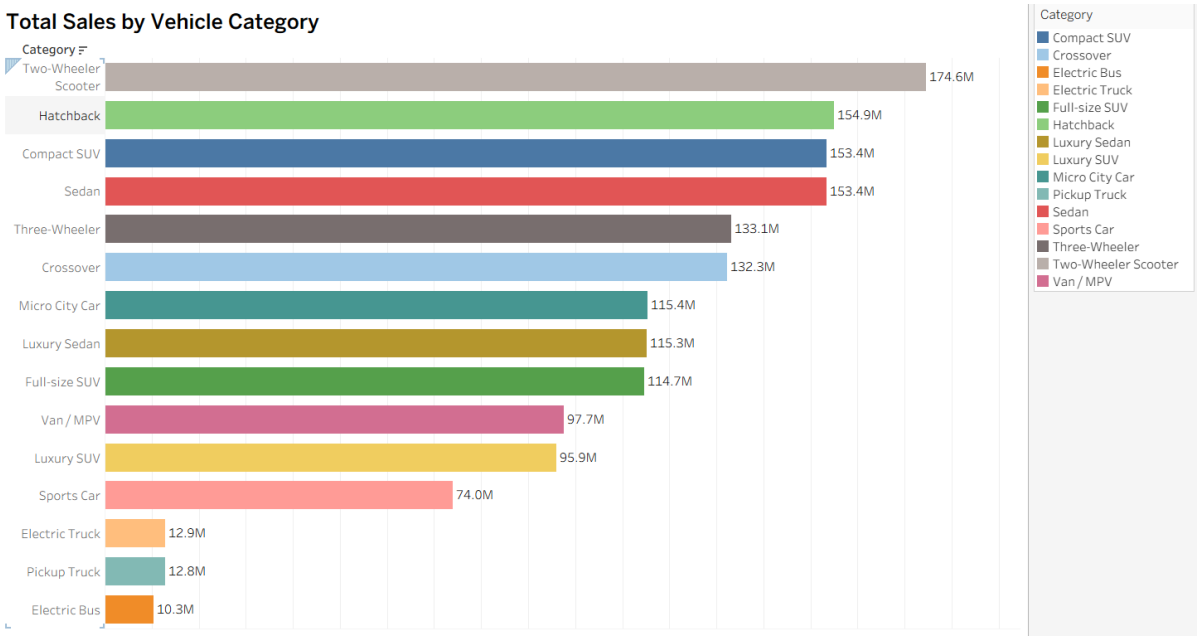
6.1 Performance Testing

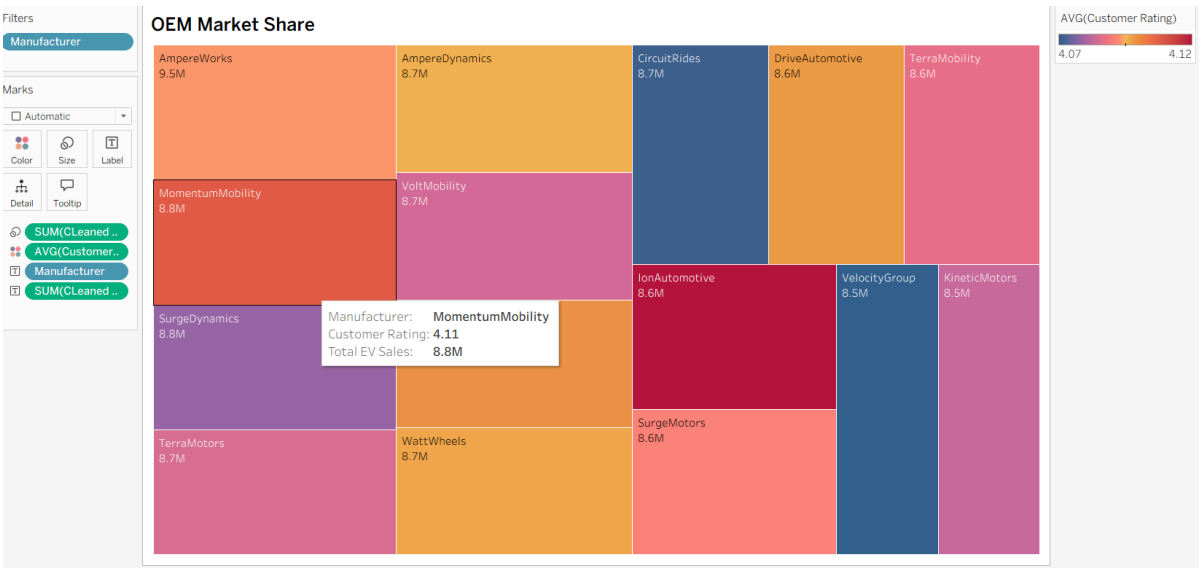
Data Rendered: Global EV Market Data loaded successfully for 1550.7M EV Sales.
Filters: Category, Battery Type, and Manufacturer filters tested and working seamlessly.
Calculations: Total EV Sales, Average Price, Median Range KPIs validated for accuracy.
Visualizations: 9 Dashboard Visuals and 6 Story Points successfully rendered.

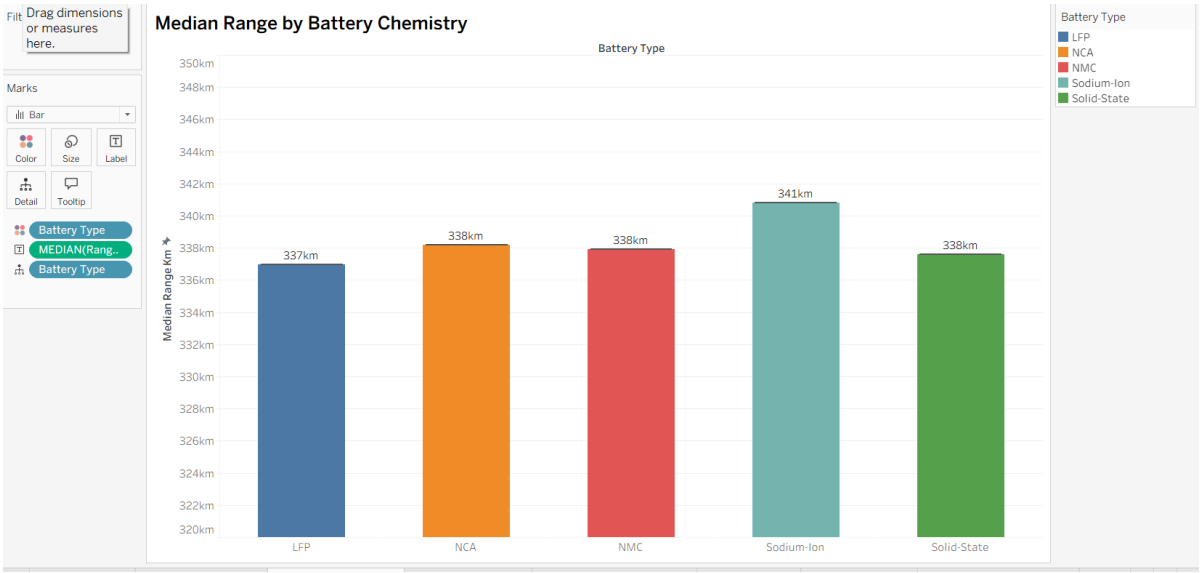
7. RESULTS

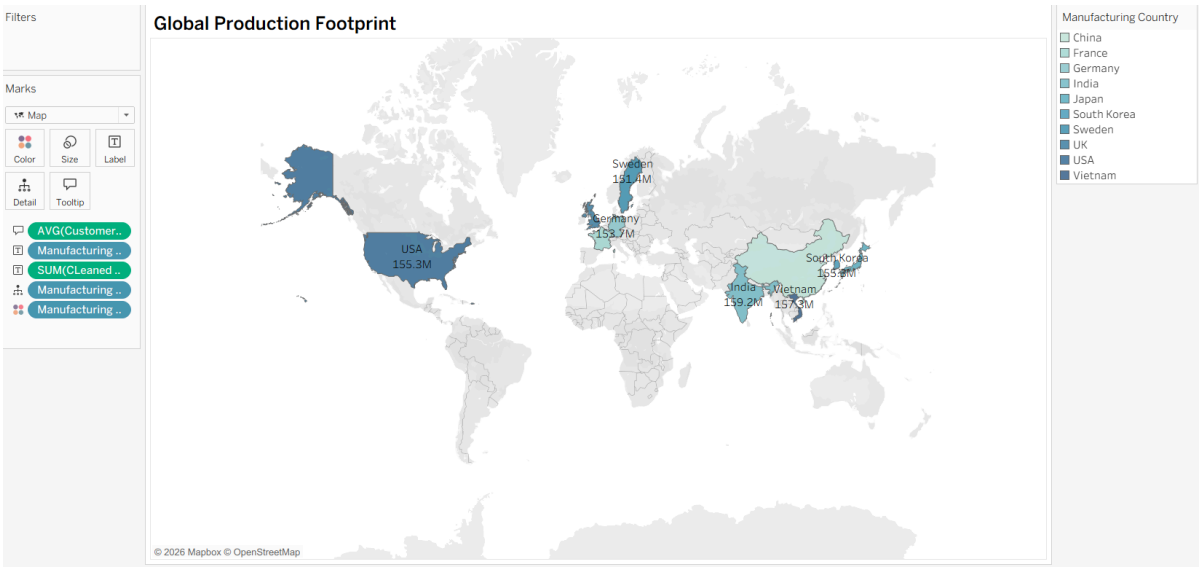
7.1 Output Screenshots

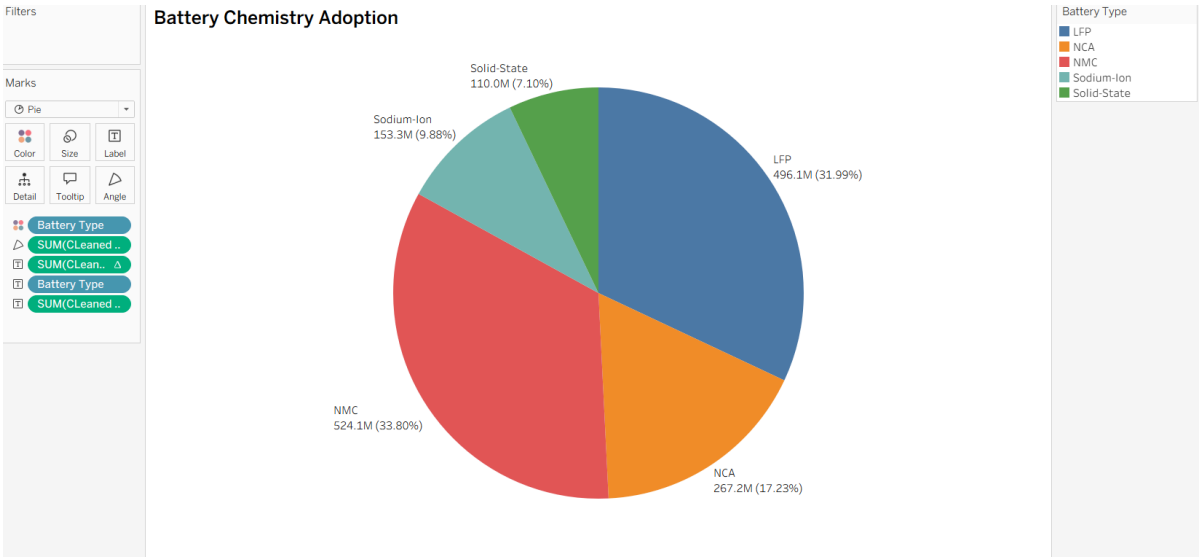


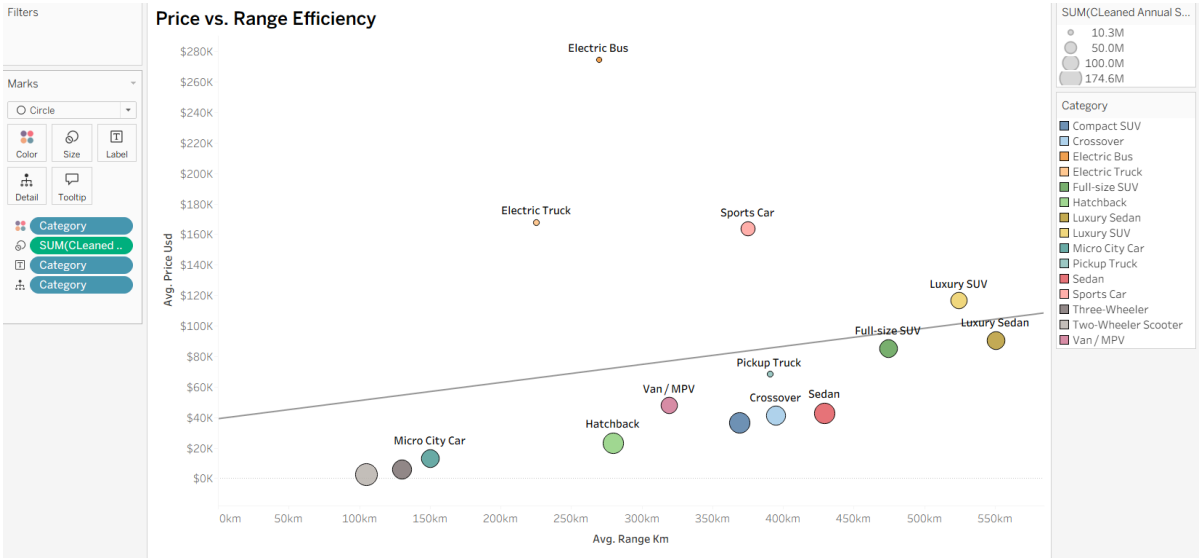


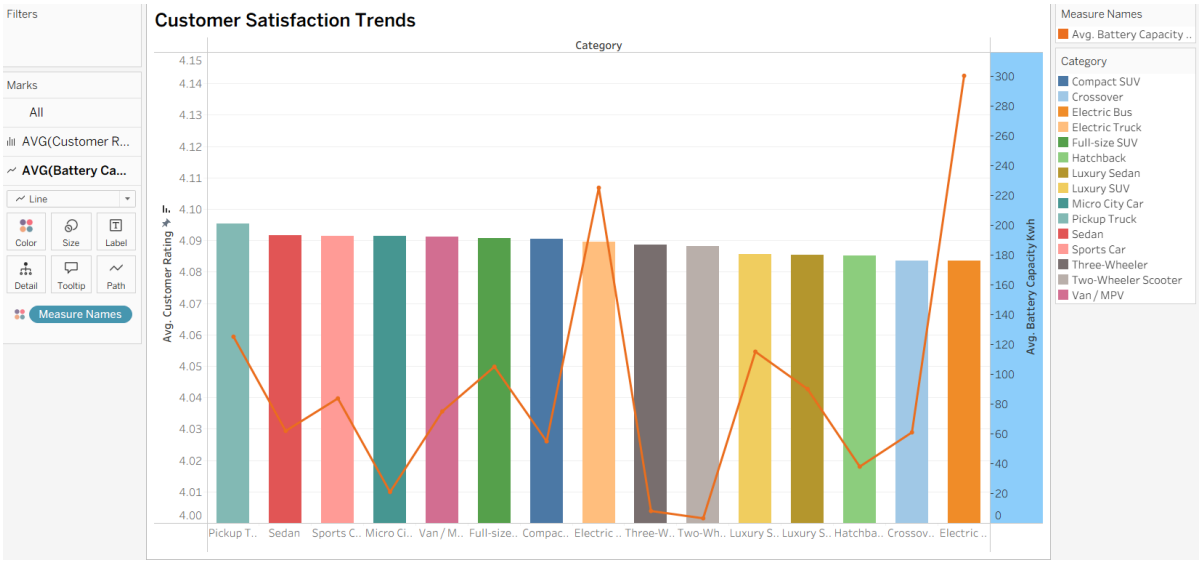

Filters
Manufacturer
Marks
Automatic
Color
Size
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Detail
Tooltip
SUM(CLeaned ..
AVG(Customer..
Manufacturer
SUM(CLeaned ..
OEM Market Share
AmpereWorks
9.5M
AmpereDynamics
8.7M
CircuitRides
8.7M
DriveAutomotive
8.6M
TerraMobility
8.6M
MomentumMobility
8.8M
VoltMobility
8.7M
IonAutomotive
8.6M
VelocityGroup
8.5M
KineticMotors
8.5M
SurgeDynamics
8.8M
Manufacturer: MomentumMobility
Customer Rating: 4.11
Total EV Sales: 8.8M
SurgeMotors
8.6M
TerraMotors
8.7M
WattWheels
8.7M
AVG(Customer Rating)
4.07
4.12

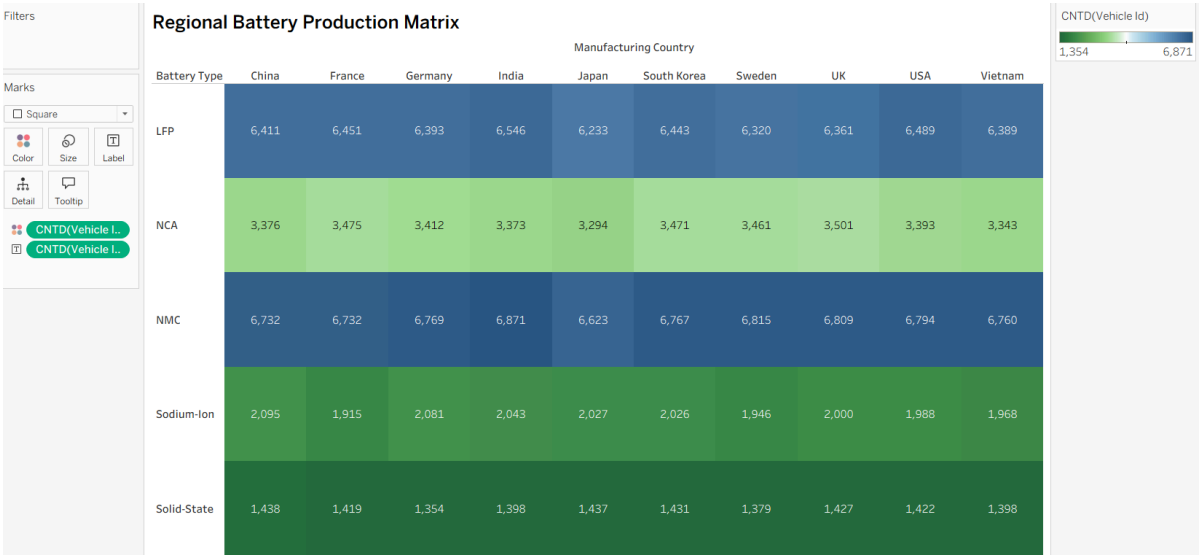

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Detail
Tooltip
Battery Type
MEDIAN(Rang..
Battery Type
Median Range by Battery Chemistry
Battery Type
Median Range Km
350km
348km
346km
344km
342km
340km
338km
336km
334km
332km
330km
328km
326km
324km
322km
320km
337km
338km
338km
341km
338km
LFP
NCA
NMC
Sodium-Ion
Solid-State
Battery Type
LFP
NCA
NMC
Sodium-Ion
Solid-State

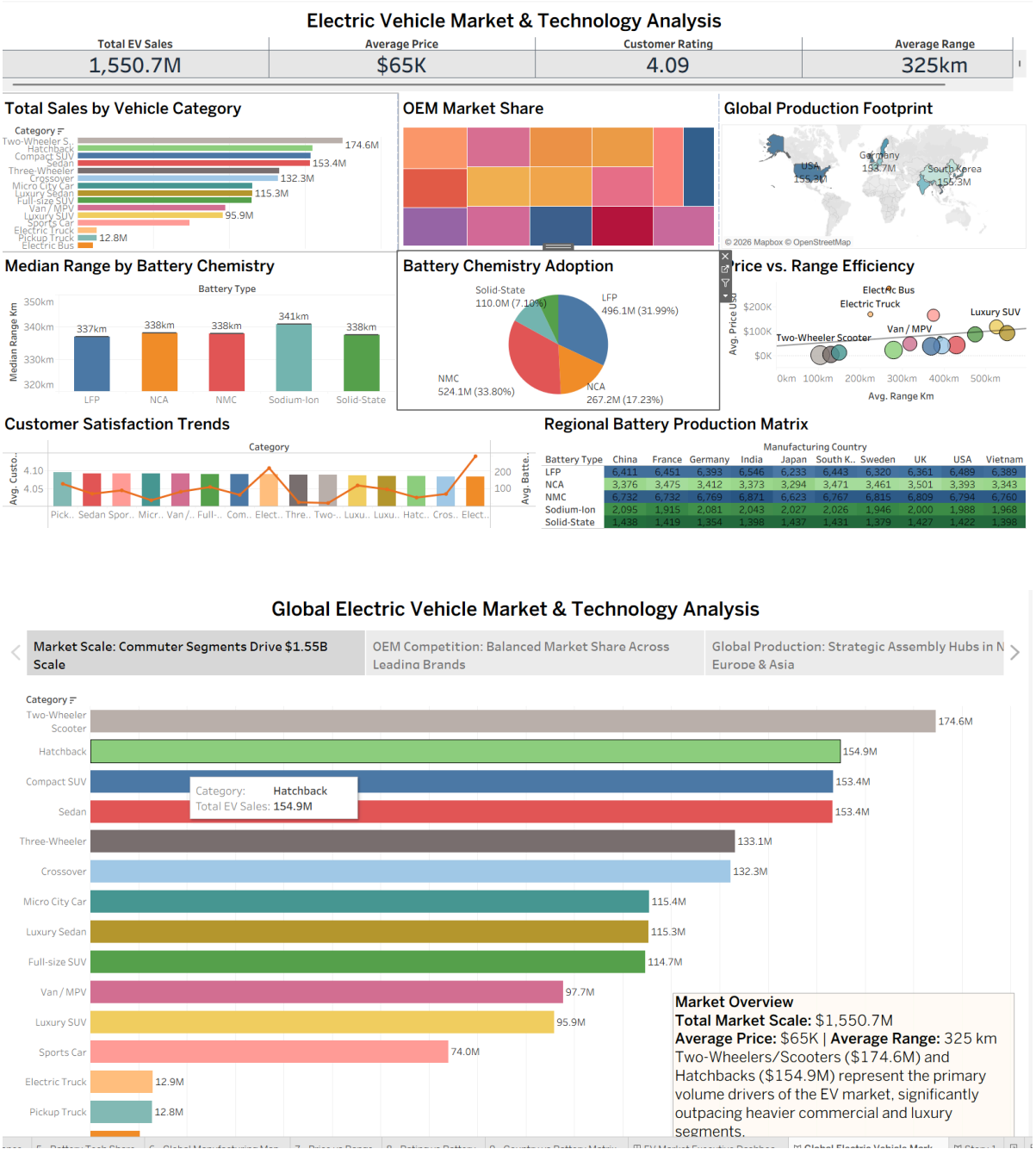

Filters
Marks
Map
Color
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Label
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Tooltip
AVG(Customer..
Manufacturing ..
SUM(CLeaned ..
Manufacturing ..
Manufacturing ..
Global Production Footprint
Sweden
151.4M
Germany
153.7M
USA
155.3M
South Korea
165.4M
India
159.2M
Vietnam
157.3M
© 2026 Mapbox © OpenStreetMap
Manufacturing Country
China
France
Germany
India
Japan
South Korea
Sweden
UK
USA
Vietnam


Filters
Marks
Pie
Color
Size
Label
Detail
Tooltip
Angle
Battery Type
SUM(CLeaned ..
SUM(CLean..
Battery Type
SUM(CLeaned ..
Battery Chemistry Adoption
Solid-State
110.0M (7.10%)
Sodium-Ion
153.3M (9.88%)
LFP
496.1M (31.99%)
NMC
524.1M (33.80%)
NCA
267.2M (17.23%)
Battery Type
LFP
NCA
NMC
Sodium-Ion
Solid-State


Filters
Marks
Circle
Color
Size
Label
Detail
Tooltip
Category
SUM(CLeaned ..
Category
Category
Price vs. Range Efficiency
Electric Bus
Electric Truck
Sports Car
Luxury SUV
Full-size SUV
Luxury Sedan
Pickup Truck
Van / MPV
Crossover
Sedan
Hatchback
Micro City Car
Avg. Price Usd
$280K
$260K
$240K
$220K
$200K
$180K
$160K
$140K
$120K
$100K
$80K
$60K
$40K
$20K
$0K
0km
50km
100km
150km
200km
250km
300km
350km
400km
450km
500km
550km
Avg. Range Km
SUM(CLeaned Annual S...
10.3M
50.0M
100.0M
174.6M
Category
Compact SUV
Crossover
Electric Bus
Electric Truck
Full-size SUV
Hatchback
Luxury Sedan
Luxury SUV
Micro City Car
Pickup Truck
Sedan
Sports Car
Three-Wheeler
Two-Wheeler Scooter
Van / MPV


Filters
Marks
All
AVG(Customer R...
AVG(Battery Ca...
Line
Color
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Tooltip
Path
Measure Names
Customer Satisfaction Trends
Category
Avg. Customer Rating
Avg. Battery Capacity Kwh
4.15
4.14
4.13
4.12
4.11
4.10
4.09
4.08
4.07
4.06
4.05
4.04
4.03
4.02
4.01
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300
280
260
240
220
200
180
160
140
120
100
80
60
40
20
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Pickup T..
Sedan
Sports C..
Micro Ci..
Van / M..
Full-size..
Compac..
Electric ..
Three-W..
Two-Wh..
Luxury S..
Luxury S..
Hatchba..
Crossov..
Electric ..
Measure Names
Avg. Battery Capacity ..
Category
Compact SUV
Crossover
Electric Bus
Electric Truck
Full-size SUV
Hatchback
Luxury Sedan
Luxury SUV
Micro City Car
Pickup Truck
Sedan
Sports Car
Three-Wheeler
Two-Wheeler Scooter
Van / MPV


Filters
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Color
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Label
Detail
Tooltip
CNTD(Vehicle I..
CNTD(Vehicle I..
Regional Battery Production Matrix
Manufacturing Country
Battery Type
China
France
Germany
India
Japan
South Korea
Sweden
UK
USA
Vietnam
LFP
6,411
6,451
6,393
6,546
6,233
6,443
6,320
6,361
6,489
6,389
NCA
3,376
3,475
3,412
3,373
3,294
3,471
3,461
3,501
3,393
3,343
NMC
6,732
6,732
6,769
6,871
6,623
6,767
6,815
6,809
6,794
6,760
Sodium-Ion
2,095
1,915
2,081
2,043
2,027
2,026
1,946
2,000
1,988
1,968
Solid-State
1,438
1,419
1,354
1,398
1,437
1,431
1,379
1,427
1,422
1,398
CNTD(Vehicle Id)
1,354
6,871



8. ADVANTAGES AND DISADVANTAGES

Advantages: Provides instant, interactive insights across multiple metrics. Web-accessible via Tableau Public without requiring local software installation. Easily readable executive summary through the 6-point Data Story.

Disadvantages: Relies on a static CSV file; requires manual updates for new data. Tableau Public hosts data publicly, so confidential corporate data cannot be used.

9. CONCLUSION

The Electric Vehicle Market Performance and Insights project successfully bridges the gap between raw, fragmented automotive data and actionable business intelligence. By leveraging Tableau, the dashboard provides stakeholders with clear correlations between battery tech, pricing, and regional production, empowering data-driven strategic decisions.

10. FUTURE SCOPE

Integrate a live database connection for automated, real-time data updates. Implement predictive analytics to forecast future EV sales based on historical trends. Expand the dataset to include detailed charging infrastructure availability by region.

11. APPENDIX

Dataset Link:
https://drive.google.com/file/d/17IihxSD__nW_tyyE1bEGwEBxvsEk1lWi/view
GitHub and Project Demo Link:
https://public.tableau.com/shared/H5JHB53HX?:display_count=n&:origin=viz_share_link

https://public.tableau.com/shared/4SZ8GJ3XC?:display_count=n&:origin=viz_share_link

https://drive.google.com/file/d/1XQNyMvwG69HMb-HVRmvU6WQnYcNUSQy3/view?usp=sharing